19MNG338 - OPERATIONS RESEARCH

Operating Room Capacity Planning and Utilization Optimization Using Surgical Workflow Analytics

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Problem Statement

- Hospital operates 8 OR suites shared across 10 surgical specialties.
- Every week, must decide how many OR-hours to give each specialty.
- Too few hours wastes capacity; too many starves other specialties and causes overtime.
- **Goal:** Optimize weekly OR-hour allocations across 10 specialties to maximize surgical throughput while controlling overtime, subject to total suite capacity and historical demand bounds.
- Must respect total OR capacity and each specialty's historical demand.
- **Key Challenge:** Throughput vs. Overtime are conflicting objectives.

Dataset Description

- **Source:** Kaggle – Optimizing Operating Room Utilization (Q1 2022)
- 2,172 real surgical case records, 8 OR suites, 10 specialties
- kaggle.com/datasets/thedevastator/optimizing-operating-room-utilization

Attribute	Purpose
index	Row identifier for the record
Encounter ID	Unique ID for the patient encounter/visit
Date	Calendar date the surgery took place
OR Suite	Identifies which operating room was used (1–8)
Service	Surgical specialty (e.g., Orthopedics, ENT)
CPT Code	Standard billing code for the procedure
CPT Description	Plain-language description of the procedure
Booked Time (min)	Time originally scheduled for the surgery
OR Schedule	Time the case was scheduled to start
Wheels In	When the patient physically arrived in the OR
Start Time	When the surgery actually began
End Time	When the surgery actually ended
Wheels Out	When the patient left the OR

Data Cleaning and Derived Metrics

- Fixed inconsistent date formats (DD-MM-YYYY vs MM/DD/YY)
- Actual Duration = (End Time − Start Time) × 1440
- Overtime = MAX(0, Actual Duration − Booked Time)
- Turnover = Next Wheels In − Current Wheels Out (same suite, same day)
- Utilization % = Total Actual Duration ÷ (Last Wheels Out − First Wheels In)
- Computed for all 497 unique OR Suite × Date combinations

Proposed Solution

- **Approach:** A five-stage Excel-based analytics and optimization pipeline, moving from raw data to a decision-ready model.
- **Clean the dataset** — standardize inconsistent date/time formats, convert text-stored numbers into real numeric fields
- **Compute derived metrics** — Actual Duration, Overtime, Turnover, and Utilization % for every OR Suite × Date combination
- **Build an on-demand statistics dashboard** — query by OR Suite or by Service to explore the data interactively
- **Formulate a Linear Programming model** — decide weekly OR-hour allocation per specialty to maximize surgical throughput, solve with Excel Solver and interpret via Sensitivity Analysis
- **Extend to Goal Programming** — balance throughput against overtime simultaneously, instead of optimizing either one alone
- **Outcome:** A data-driven OR scheduling recommendation grounded in one quarter of real surgical records, with a clear, interpretable trade-off between throughput and overtime.

Statistical Analysis

Built an on-demand statistics dashboard in Excel — enter a value, get instant summary statistics.

View by OR Suite — type a suite number (1–8), instantly see:

- Average utilization %, Total actual surgery minutes used, Average daily duration, Number of active days in the quarter

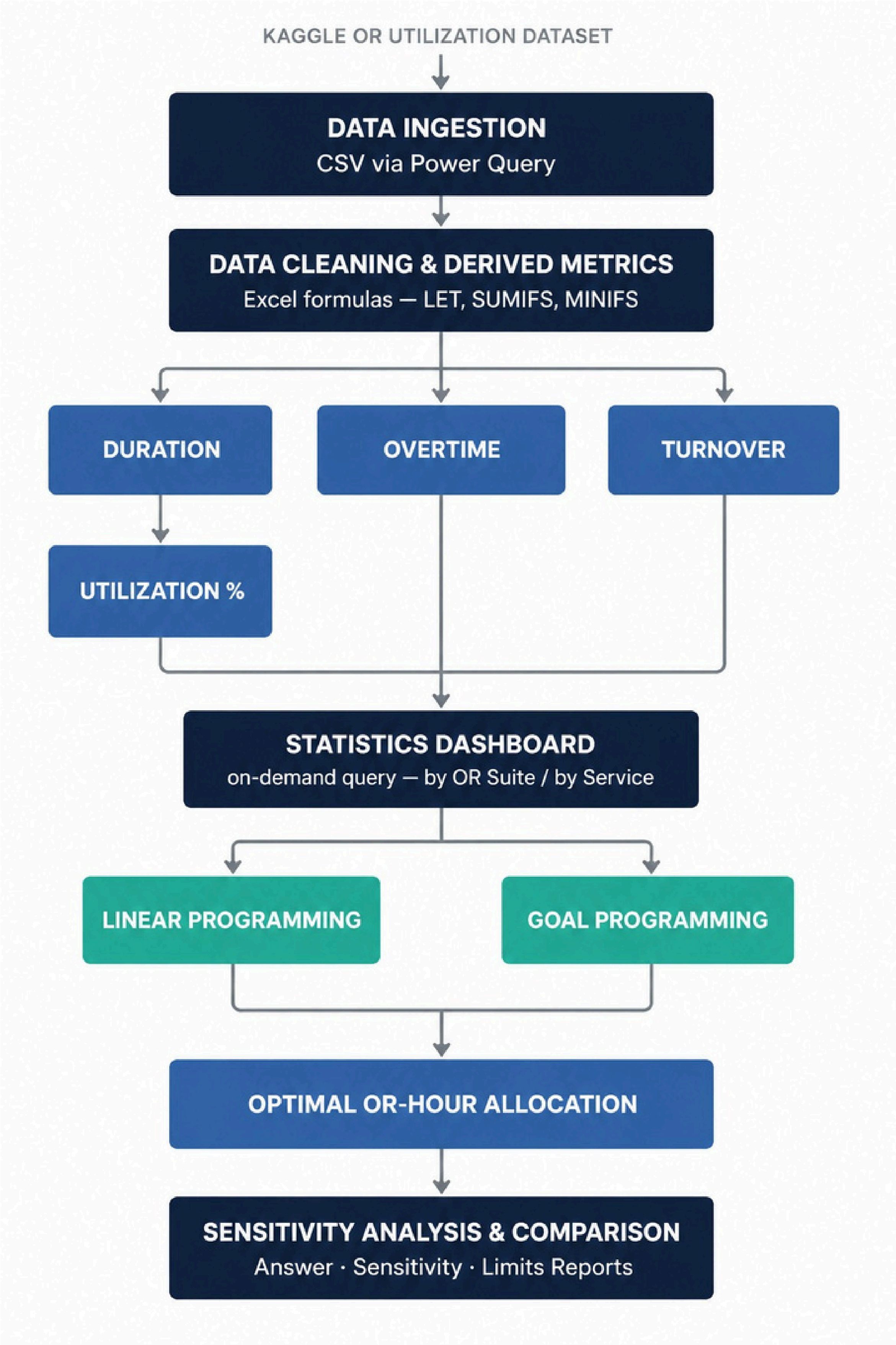
View by Service — type a specialty name, instantly see:

- Number of surgeries performed, Average actual duration per case, Average overtime per case, Average booked time per case

Key Finding:

- Average OR utilization across all 8 suites: ~46%
- Indicates significant unused OR capacity across the hospital
- This finding directly motivates the optimization models that follow

Methodology



Linear Programming Model

Motivation: Before balancing multiple goals, first find the single best allocation that maximizes surgical throughput alone — this becomes the baseline the Goal Programming model builds on.

Decision Variable: X_s = OR-hours allocated to specialty s per week (s = 1 to 10)

Objective Function: Maximize $Z = \Sigma\ (60 \times X_s \div D_s)$

- X_s : OR-hours given to specialty s
- D_s : average actual case duration for specialty s (minutes)
- $60 \div D_s$ converts hours into number of cases performed

Constraints:

- $\Sigma\ X_s \leq 320$ (total weekly OR capacity, all 8 suites combined)
- $0.8 \times$ historical average $\leq X_s \leq 1.2 \times$ historical average (stay within realistic demand per specialty)
- $X_s \geq 0$ (non-negativity)

Goal Programming Model

Motivation: Throughput maximization alone ignores overtime; Goal Programming balances both conflicting objectives.

Goals:

Goal 1 (Throughput Target): $\Sigma\ (60 \times X_s \div D_s) + d1^- - d1^+ = 200$ cases/week

Goal 2 (Overtime Control): $\Sigma\ \text{Overtime}_s + d2^- - d2^+ =$ Target Overtime

Objective Function: Minimize $Z = W1 \times d1^- + W2 \times d2^+$

- $d1^-$: Under-achievement penalty for case throughput
- $d2^+$: Over-achievement penalty for exceeding overtime limits
- W1, W2 : Decision-maker priority weights

Hard Constraints:

- $\Sigma\ X_s \leq 320$ (Total weekly capacity)
- $0.8 \times \text{avg} \leq X_s \leq 1.2 \times \text{avg}$ (Specialty demand bounds)
- $X_s, d1^-, d1^+, d2^-, d2^+ \geq 0$

Tools and Significance

Tools Used:

- Microsoft Excel — primary platform for data, models, and dashboard
- Power Query — importing and cleaning the raw CSV dataset

Significance:

- **Capacity Insight** — Sensitivity Analysis reveals the true bottleneck is specialty demand caps, not OR capacity (only 152 of 320 hours used)
- **Overtime Awareness** — Goal Programming explicitly controls overtime instead of ignoring it, unlike single-objective LP
- **Data-Driven Scheduling** — allocation decisions are grounded in one quarter of real historical surgical records, not guesswork
- **Multi-Objective Decision Making** — hospital administrators can weigh throughput against overtime instead of being locked into one goal

Project Status

Completed:

- Dataset identified and loaded — Kaggle OR Utilization dataset, Q1 2022
- Data cleaning completed — standardized date/time formats, computed derived metrics (Duration, Overtime, Turnover, Utilization %)

Current Status:

- Formulating the Linear Programming model — defining decision variables, objective function, and constraints for weekly OR-hour allocation

Future Work:

- For larger datasets or multi-quarter analysis, migrate the aggregation and modeling layer to Python (Pandas for data processing, PuLP/SciPy for optimization) while keeping the same underlying LP/Goal Programming formulation.